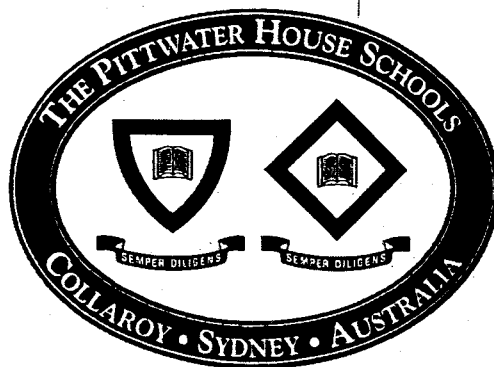


NAME: _____

STUDENT No: _____



YEAR 12 TRIAL HSC EXAMINATION 2004

MATHEMATICS

*Time Allowed: 3 hours
(plus 5 minutes reading time)*

INSTRUCTIONS FOR CANDIDATES

- Start each question in a new booklet.
- All questions are to be answered.
- All questions are of equal value.
- The marking scheme has been given on the right hand side of the page.
- Standard integrals are printed on the back page.
- Approved calculators may be used.

Question 1.
(Start a new booklet)

- | | Marks |
|--|--------------|
| (a) $ 3 - x = 7$ | 2 |
| (b) Differentiate $y = 2\sqrt{x} + \cos x$ with respect to x . | 2 |
| (c) An arc of length 8 cm subtends an angle of θ at the centre of a circle with radius 3 cm. Find θ to the nearest degree. | 2 |
| (d) Differentiate $f(x) = \ln\left(\frac{x+1}{x-1}\right)$. Answer in simplest form. | 2 |
| (e) Evaluate $e^{-2.6}$ correct to 3 significant figures. | 2 |
| (f) At a sale, 30% discount is given off the marked price. If Anna pays \$175 for a coat at this sale, what was the original marked price? | 2 |

Question
(Start a

(a) Fi

(b) Li

Li

(i)

(ii)

(iii)

(iv)

(v)

(vi)

(vii)

Question 2.
(Start a new booklet)

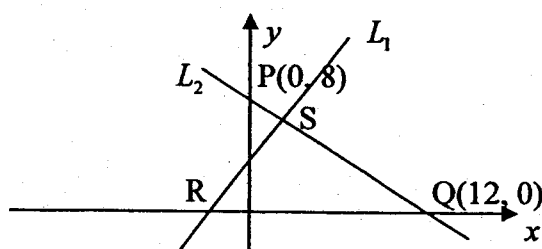
Marks

- (a) Find the equation of the normal to $y = 2e^{2x}$ at the point $x = 0$

3

- (b) Line L_1 has the equation $3x - 2y + 3 = 0$

Line L_2 passes through $P(0, 8)$ and $Q(12, 0)$



- (i) L_1 cuts the x axis at R . Find the coordinates of R .
- (ii) Find the gradient of line L_2
- (iii) Show L_1 and L_2 are perpendicular
- (iv) Show the equation of L_2 is $2x + 3y - 24 = 0$
- (v) Find S , the point of intersection of L_1 and L_2
- (vi) Find the area of ΔRSQ
- (vii) Copy the diagram and shade the region where $2x + 3y - 24 \leq 0$, $3x - 2y + 3 \geq 0$ and $y \geq 0$ hold simultaneously.

Question 3.
(Start a new booklet)

Marks

Question
(Start a new booklet)

(a) Find $\int \frac{2x}{x^2 - 4} dx$

1

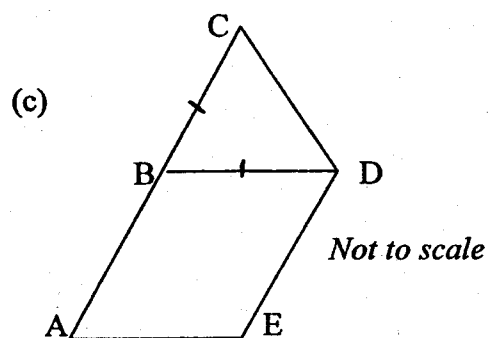
(a) A p
of 2

At
trav
tow

(b) Evaluate $\int_0^{\frac{\pi}{2}} 1 + \cos x dx$

2

(i)



A, B and C are collinear

ABDE is a parallelogram

$\triangle BCD$ is isosceles with $BC = BD$

$\angle BCD = 58^\circ$

(ii)

(b) A p

(i)

(ii)

(i) Copy the diagram

(ii) Find the size of $\angle AED$ giving reasons

3

(iii)

(iv)

(d) Differentiate with respect to x

2

(v)

(i) $(3x^2 - 5)^7$

(ii) $x^3 \tan x$

(c) The c

(i)

(ii)

(e) $\int_0^1 (3x+1)^3 dx$

2

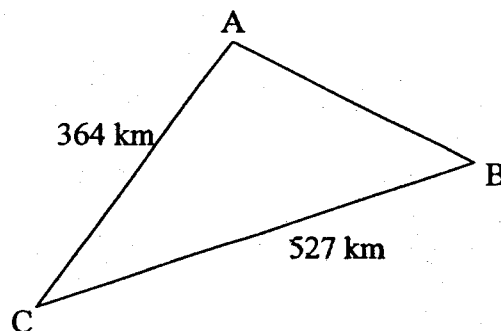
Question 4.

(Start a new booklet)

Marks

- (a) A plane is flying from town A on a bearing of 237°T to town C which is 364 km away.

At town C, the plane changes its course and travels 527 km on a bearing of 077°T to town B.



- (i) Copy the diagram and show $\angle ACB = 20^\circ$

2

- (ii) Find the distance of B from A. Answer to the nearest km.

2

- (b) A parabola has the point (2, 1) as its focus and its directrix is $y = -3$. Write down:

5

- (i) the vertex

- (ii) the equation of its axis

- (iii) the focal length

- (iv) the equation of the parabola; and

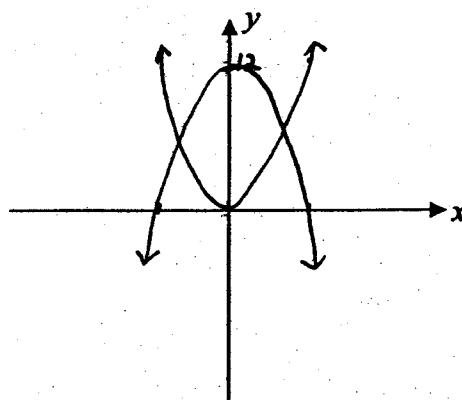
- (v) draw its sketch.

- (c) The curves $y = 2x^2$ and $y = 12 - x^2$ are drawn

3

- (i) Show the curves intersect at (-2, 8) and (2, 8)

- (ii) Hence, find the area between the curves



Question 5.
(Start a new booklet)

Marks

- (a) The first 3 terms of a geometric series are $x - 2$, $x + 1$, $3x - 3$.
- (i) Find the 2 values of x
- (ii) Write down the 2 series and find their ratios
- (iii) Explain why these series don't have a limiting sum
- (b) The gradient function of a curve is given by $3x^2 - 11$. If the curve passes through (3, 4) find its equation.
- (c) Find k if $4x^2 - kx + 1 = 0$ has real roots.
- (d) Solve $2\sin x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$

2

1

1

3

3

2

Question 6.
(Start a new booklet)

(a) Solve the equation

(b) A curve is defined by the equation

Determine the limiting sum of the series

Consider the function

(i) Find the gradient of the curve at the point

(ii) Find the equation of the normal to the curve at the point

(iii) Find the area of the region bounded by the curve, the x-axis and the line

(iv) Find the volume of the solid generated when the region is rotated about the y-axis

(c) A particle moves in a straight line with constant acceleration. It starts from rest and reaches a speed of 10 m/s after 5 seconds.

(i) Find the acceleration

(ii) Find the distance travelled

(d) A particle moves in a straight line with constant acceleration. It starts from rest and reaches a speed of 10 m/s after 5 seconds.

Question 6.
(Start a new booklet)

Marks

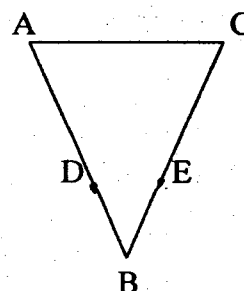
(a) Solve $\log_3(2x - 4) = 4$

2

(b) ABC is a triangle with $AB = BC$.

D and E are points of AB and CB such that $AD = CE$.

Copy the diagram.



(i) Why does $DB = EB$?

1

(ii) Prove $\triangle ABE$ is congruent to $\triangle CBD$.

2

(iii) Why does $\angle BAE = \angle BCD$?

1

(iv) Prove $\angle CAE = \angle ACD$.

1

(c) A point P (x, y) moves so that its distance from A(3, 0) is twice its distance from B(0, 3).

(i) Show the equation of the locus of P is $x^2 + 2x + y^2 - 8y + 9 = 0$

2

(ii) Show this locus is a circle and write down its centre and radius.

2

(d) Change $\frac{3\pi}{5}$ radians to degrees.

1

Question 7.
(Start a new booklet)

Marks

(a) For the curve $y = \frac{1}{3}x^3 - x^2 - 8x + 12$,

- (i) Find any turning points.
- (ii) Determine their nature
- (iii) Find any points of inflexion
- (iv) Sketch the curve showing all relevant features
- (v) For what values of x is the curve concave up?
- (vi) When is the curve decreasing?

2

1

2

1

1

1

(b) An arithmetic progression has a first term of 1 and a last term of 14. If the sum of the series is 90,

- (i) Find the number of terms in the series
- (ii) Show the difference is $\frac{13}{11}$
- (iii) Find an expression for the n th term as a single fraction.

2

1

1

Question
(Start a new booklet)

(a) Find

(b) Es

Ar

(c) So

(d) Wh
The
at t

(i)

(ii)

(iii)

Question 8.

(Start a new booklet)

Marks

- (a) Find A , B and C if $3x^2 + 4x + 2 \equiv Ax(x-1) + B(x+1) + C$ 3
- (b) Estimate $\int_0^1 \sin(1+x^2) dx$ using Simpson's Rule with 3 function values.
Answer to three decimal places. 3
- (c) Solve $2\sin^2 x - 3\sin x + 1 = 0$ for $0 \leq x \leq 2\pi$ 2
- (d) When Tom started university, his parents borrowed \$30000 to pay for his education. They repaid the loan by making 48 equal monthly repayments. Interest was charged at the rate of 1.1% per month on balance owing.
- (i) Write an expression for the amount owing after 1 monthly repayment of \$ m . 1
- (ii) Show that after 2 monthly repayments, the amount owing is $\$30663.63 - 2.011m$. 1
- (iii) Calculate the value of each monthly repayment. 2
Answer to the nearest cent.

Question 9.

(Start a new booklet)

Marks

(a) If α and β are roots of $3x^2 + 5x - 2 = 0$, find:

(i) $\alpha + \beta$

1

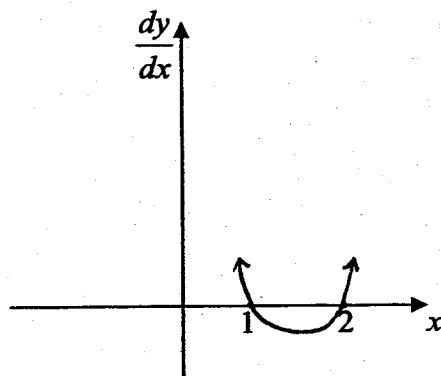
(ii) $\alpha\beta$

1

(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

2

(b) The sketch shows the graph of $\frac{dy}{dx}$ which is the gradient function of $y = f(x)$.

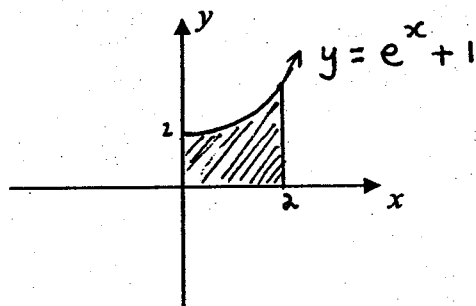


If $f(0) = 1$, draw a possible sketch of $y = f(x)$.

2

(c) What is the volume of the solid formed when the shaded area is rotated about the x axis?

3



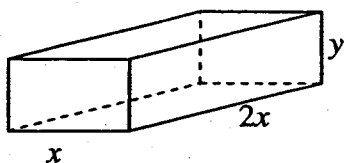
(d) Solve $\frac{1}{2}(5x-1) = 4 - \frac{1}{3}(x-2)$

3

Question 10.
(Start a new booklet)

Marks

- (a) Boxes in the shape of rectangular prisms are to be made so that the width (x metres) of the base is half the length.



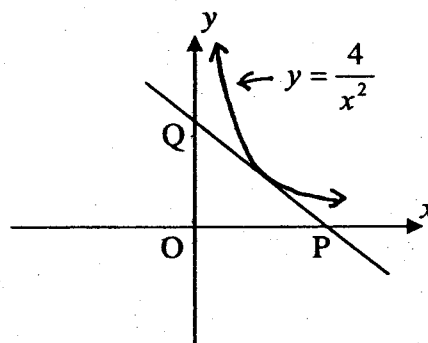
Each box holds a volume of 4 cubic metres. The wood used to make the top and bottom of the box costs \$15 per m^2 while the wood for the other 4 sides costs \$10 per m^2 .

- (i) Find an expression for y in terms of x . 1
- (ii) Show the cost \$ C of building each box is given by $C = 60x^2 + \frac{120}{x}$. 2
- (iii) What are the measurements of the cheapest boxes that can be constructed? 3

- (b) PQ is a tangent to $y = \frac{4}{x^2}$.

It cuts the x axis at P and the y axis at Q.

$$\angle OPQ = \angle OQP$$



- (i) Explain why the gradient of PQ is -1. 1
- (ii) Show PQ is a tangent touching the curve at the point (2, 1). 2
- (iii) Find the equation of this tangent. 1
- (c) Differentiate $y = \cot x$. Answer in the simplest form. 2

Question 1

$$|3-x| = 7$$

$$3-x = 7 \quad \text{or} \quad 3-x = -7$$

$$x = -4 \quad \text{or} \quad x = 10$$

$$y = 2\sqrt{x} + \cos x = 2x^{\frac{1}{2}} + \cos x$$

$$\frac{dy}{dx} = x^{-\frac{1}{2}} - \sin x$$

$$= \frac{1}{\sqrt{x}} - \sin x$$

$$l = 8 \quad r = 3$$

$$l = 10 \quad \therefore 8 = 30$$

$$\theta = \frac{8}{3} \text{ radians}$$

$$= \frac{8}{3} \times \frac{180}{\pi}$$

$$= 152^{\circ} 47'$$

$$= 153^{\circ} \text{ (to nearest deg.)}$$

$$f(x) = \ln\left(\frac{x+1}{x-1}\right) = \ln(x+1) - \ln(x-1)$$

$$f'(x) = \frac{1}{x+1} - \frac{1}{x-1} = \frac{x-1-(x+1)}{(x+1)(x-1)}$$

$$= \frac{x-1-x-1}{x^2-1} = \frac{-2}{x^2-1}$$

$$-2.6$$

$$= 0.074273578$$

$$= 0.0743 \text{ to 3 sig. figs.}$$

$$70\% = \$175$$

$$1\% = \frac{175}{70} = 2.5$$

$$100\% = 250$$

$$\therefore \text{Original price} = \$250$$

Question 2

$$y = 2e^{2x}$$

$$y = 2e^{2x} \times 2 = 4e^{2x}$$

$$\text{At } x=0, \frac{dy}{dx} = 4 \quad y = 2$$

$$\therefore m=4 \quad P=(0,2) \quad m_{\text{normal}} = -\frac{1}{4}$$

Eqn of normal is

$$y-2 = -\frac{1}{4}(x-0)$$

$$4y-8 = -x$$

$$x+4y-8=0$$

i) L_1 is $3x-2y+3=0$

Cuts x axis when

$$y=0 \quad \text{ie } 3x+3=0$$

$$x=-1$$

$$\therefore R = (-1,0)$$

$$ii) L_2 \text{ has } m = \frac{8-0}{0-12} = -\frac{2}{3}$$

$$iii) 3x-2y+3=0$$

$$2y = 3x+3$$

$$y = \frac{3}{2}x + \frac{3}{2}$$

$$m = \frac{3}{2}$$

$$m_{L_1} \times m_{L_2} = \frac{3}{2} \times -\frac{2}{3} = -1$$

$\therefore L_1$ and L_2 are $\perp$

iv) Equation of L_2 is

$$y-8 = -\frac{2}{3}(x-0)$$

$$3y-24 = -2x$$

$$2x+3y-24=0$$

$$v) 3x-2y=-3 \quad (1)$$

$$2x+3y=24 \quad (2)$$

$$\times (1) \text{ by } 3$$

$$9x-6y=-9 \quad (3)$$

$$\times (2) \text{ by } 2$$

$$4x+6y=48 \quad (4)$$

$$(3)+(4)$$

$$13x=39$$

$$x=3, y=6$$

$$\therefore S = (3,6)$$

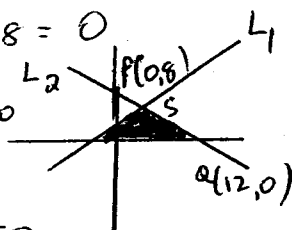
$$vi) R=(-1,0) \quad Q=(12,0)$$

$$\therefore RQ = 13 \quad \text{Height is } 6$$

$$\text{Area} = \frac{1}{2} \times 13 \times 6$$

$$= 39 \text{ u}^2$$

vii) See diagram.



Question 3

$$\int \frac{2x}{x^2-4} dx = \ln(x^2-4) + C$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} 1 + \cos x dx &= \left[x + \sin x \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{2} + \sin \frac{\pi}{2} - (0 + \sin 0) \\ &= \frac{\pi}{2} + 1 \end{aligned}$$

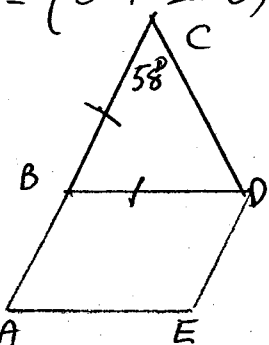
i) 1)

$$\angle BDC = \angle BCD = 58^\circ$$

Base L's of isos. Δ) A

$$\angle ABD = \angle BDC + \angle BCD = 58^\circ + 58^\circ = 116^\circ$$

$$\angle AED = 116^\circ (\text{opp. L's of parm. } =)$$



$$\begin{aligned} \frac{d}{dx} (3x^2-5)^7 &= 7(3x^2-5)^6 \times 6x \\ &= 42x(3x^2-5)^6 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} x^3 + \tan x &= x^3 \sec^2 x + \tan x \times 3x^2 \\ &= x^2(x \sec^2 x + 3 \tan x) \end{aligned}$$

$$\begin{aligned} \int_0^1 (3x+1)^3 dx &= \left[\frac{(3x+1)^4}{4 \times 3} \right]_0^1 \\ &= \frac{256}{12} - \frac{1}{12} \\ &= \frac{255}{12} \text{ or } 21\frac{1}{4} \end{aligned}$$

Question 4

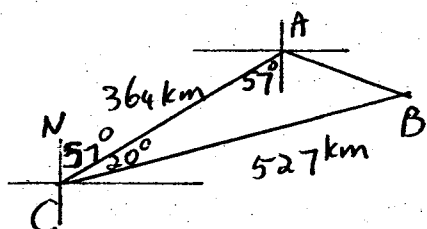
$$\angle NCA = 57^\circ$$

(alt L's =)

$$\angle NCB = 77^\circ$$

$$= 57 + 20$$

$$\therefore \angle ACB = 20^\circ$$



$$\begin{aligned} AB^2 &= 364^2 + 527^2 - 2 \times 364 \times 527 \times \cos 20^\circ \\ &= 49706.28788 \end{aligned}$$

$$AB = 222.949$$

$$= 223 \text{ km to nearest km}$$

then x

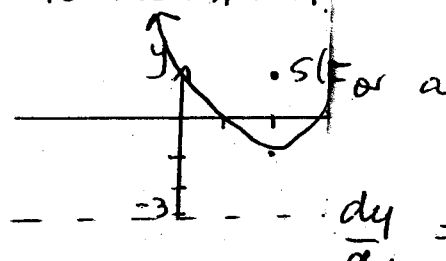
$$b) i) V = (2, -1)$$

$$ii) \text{axis is } x=2$$

$$iii) a=2$$

$$iv) (x-2)^2 = 8(y+1)$$

v) See sketch



$$\begin{aligned} c) i) y &= 2x^2 \text{ and } y = 12 - x^2 \\ \text{intersect when } 2x^2 &= 12 - x^2 \end{aligned}$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = 2$$

$$\text{or } x = -2$$

$$y = 8$$

$$y = 8$$

$$\begin{aligned} ii) \text{Area} &= 2 \int_0^2 (12 - x^2 - 2x^2) dx \\ &= 2 \int_0^2 (12 - 3x^2) dx \\ &= 2 \left[12x - x^3 \right]_0^2 \\ &= 2(24 - 8) \\ &= 32 \text{ u}^2 \end{aligned}$$

Question 5

$$a) x-2, x+1, 3x-3$$

$$i) \frac{x+1}{x-2} = \frac{3x-3}{x+1}$$

$$x^2 + 2x + 1 = 3x^2 - 9x + 6$$

$$2x^2 - 11x + 5 = 0$$

$$(2x-1)(x-5) = 0$$

$$x = \frac{1}{2} \text{ or } 5$$

Question 6

$$g_3(2x)$$

$$2x$$

$$2$$

(3) When $x = \frac{1}{2}$, Series is $-\frac{3}{2}, \frac{3}{2}, -\frac{3}{2}$
ratio is -1

When $x = 5$, Series is $3, 6, 12$
ratio is 2

For a limiting sum, $|r| < 1$.

$$\frac{dy}{dx} = 3x^2 - 11$$

$$y = x^3 - 11x + C$$

Curve passes thru $(3, 4)$

$$\therefore 4 = 27 - 33 + C$$

$$C = 10$$

$$\text{Curve is } y = x^3 - 11x + 10$$

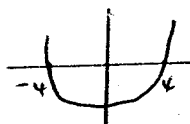
$4x^2 - kx + 1 = 0$ has real

$$\text{roots } \therefore \Delta \geq 0$$

$$k^2 - 16 \geq 0$$

$$(k+4)(k-4) \geq 0$$

$$k \leq -4 \text{ or } k \geq 4$$



$$2 \sin x = -\sqrt{3} \text{ for } 0 \leq x \leq 2\pi$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

x is in quadrants 3 or 4

$$x = \pi + \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3} \text{ or } \frac{5\pi}{3}$$

Q6

$$9(2x-4) = 4$$

$$2x-4 = \frac{4}{9}$$

$$2x-4 = 81$$

$$2x = 85$$

$$x = 42.5$$

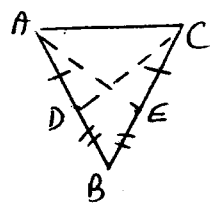
$$b) i) AB = AD + DB$$

$$CB = CE + EB$$

$$\text{But } AB = CB$$

$$\therefore AD + DB = CE + EB$$

$$\text{But } AD = CE, \therefore DB = EB$$



$$ii) AB = BC \text{ given.}$$

$$EB = DB \text{ (proven above)}$$

$\angle B$ is common.

$$\therefore \triangle ABE \equiv \triangle CBD \text{ (SAS)}$$

$$iii) \angle BAE = \angle BCD \text{ (corresp. } \angle \text{'s of } \triangle \text{'s)}$$

$$iv) \angle BAC = \angle BCA \text{ (base } \angle \text{'s of } \triangle \text{'s)}$$

$$\angle BAE + \angle CAE = \angle BCD + \angle ACD$$

$$\text{But } \angle BAE = \angle BCD \text{ (proven above)}$$

$$\therefore \angle CAE = \angle ACD$$

$$c) P(x, y) \quad A(3, 0) \quad B(0, 3)$$

$$i) PA = 2PB$$

$$\therefore (PA)^2 = 4(PB)^2$$

$$(x-3)^2 + y^2 = 4[x^2 + (y-3)^2]$$

$$x^2 - 6x + 9 + y^2 = 4x^2 + 4y^2 - 24y + 36$$

$$3x^2 + 3y^2 + 6x - 24y + 27 = 0$$

$$x^2 + y^2 + 2x - 8y + 9 = 0$$

$$ii) x^2 + 2x + 1 + y^2 - 8y + 16 = -9 + 16$$

$$(x+1)^2 + (y-4)^2 = 8$$

This is a circle centre $(-1, 4)$

$$\text{radius} = 2\sqrt{2}$$

$$d) \frac{3\pi}{5} = \frac{3 \times 180}{5} = 108^\circ$$

Question 7

$$y = \frac{1}{3}x^3 - x^2 - 8x + 12$$

$$\frac{dy}{dx} = x^2 - 2x - 8$$

For turning pt, $\frac{dy}{dx} = 0$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$x = 4 \quad \text{or} \quad x = -2$$

$$y = -14\frac{2}{3} \quad y = 21\frac{1}{3}$$

$$\frac{d^2y}{dx^2} = 2x - 2$$

$$\text{When } x = 4, \frac{d^2y}{dx^2} > 0$$

$\therefore$ min. at $(4, -14\frac{2}{3})$

$$\text{When } x = -2, \frac{d^2y}{dx^2} < 0$$

$\therefore$ max. at $(-2, 21\frac{1}{3})$

i) For a pt. of inflexion,

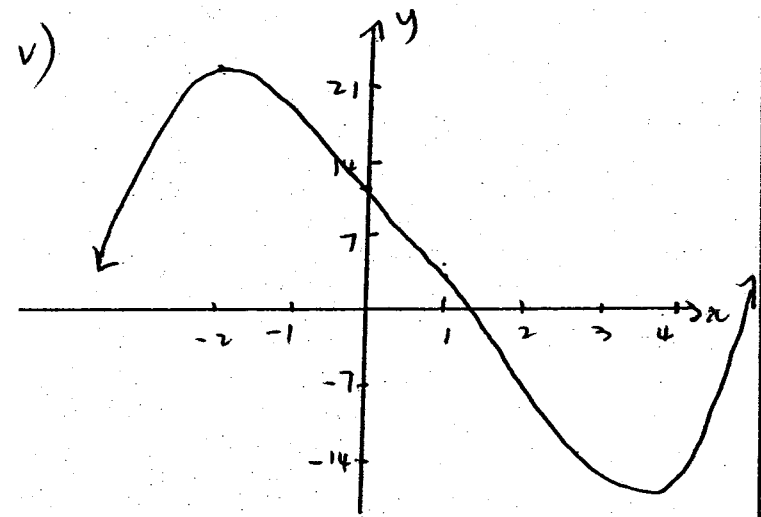
$$\frac{d^2y}{dx^2} = 0 \quad \text{ie } 2x - 2 = 0$$

$$x = 1$$

$$y = 3\frac{1}{3}$$

When $x < 1$ $\frac{d^2y}{dx^2} < 0$ } Change in
 $x > 1$ $\frac{d^2y}{dx^2} > 0$ } Concavity

$\therefore$ Pt of inflexion at $(1, 3\frac{1}{3})$

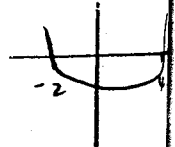


v) Curve is concave up
 for $\frac{d^2y}{dx^2} > 0$ ie $2x - 2 > 0$
 $x > 1$

vi) Curve is decreasing when

$$\frac{dy}{dx} < 0 \quad \text{ie } (x-4)(x+2) < 0$$

$$-2 < x < 4$$



b) 1, ..., 14

$$90 = \frac{n}{2}(1 + 14)$$

$$180 = 15n$$

$$n = 12 \quad \therefore 12 \text{ terms}$$

$$ii) 14 = 1 + 11d$$

$$13 = 11d$$

$$d = \frac{13}{11}$$

$$iii) t_n = a + (n-1)d$$

$$= 1 + (n-1)\frac{13}{11}$$

$$= \frac{11 + 13n - 13}{11}$$

$$= \frac{13n - 2}{11}$$

Question 8

$$a) 3x^2 + 4x + 2 \equiv Ax(x-1) + B(x+1)$$

$$3x^2 + 4x + 2 \equiv Ax^2 - Ax + Bx + B$$

$$\therefore 3 = A$$

$$4 = -A + B$$

$$4 = -3 + B$$

$$B = 7$$

$$2 = B + C$$

$$2 = 7 + C$$

$$C = -5$$

$$\therefore A = 3, B = 7, C = -5$$

1
 $\sin$
 $\frac{1-1}{2}$
 $\sin$
 $= \frac{1}{6}$
 $= \frac{1}{6}$
 $\div$
 $2 \sin$
 $(2 \sin$
 $\sin x$
 $x =$
 $x =$
 i) Am
 300
 300
 Amt.
 3000
 30663
 Amt.
 30000
 0000
 719.9
 m
 Repa

Q4) $\int_0^1 \sin(1+x^2) dx$

$$\frac{1-0}{2} = \frac{1}{2}$$

$$\sin(1+x^2) = \frac{1}{3} \times \frac{1}{2} \left[f(0) + 4f\left(\frac{1}{2}\right) + f(1) \right]$$

$$= \frac{1}{6} \left[\sin 1 + 4 \sin 1.25 + \sin 2 \right]$$

$$= \frac{1}{6} [5.546]$$

$$\approx 0.924$$

$$2 \sin^2 x - 3 \sin x + 1 = 0$$

$$(2 \sin x - 1)(\sin x - 1) = 0$$

$$\sin x = \frac{1}{2} \quad \text{or} \quad \sin x = 1$$

$$x = \frac{\pi}{6} \text{ or } \pi - \frac{\pi}{6} \text{ or } \frac{\pi}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$$

Amount owing after 1 mth

$$30000 \left(1 + \frac{1.1}{100}\right) - m$$

$$30000(1.011) - m$$

Amt. owing after 2 months

$$[30000(1.011) - m] 1.011 - m$$

$$3(x+1) + B + 030000(1.011)^2 - m(1 + 1.011)$$

$$30663.63 - 2.011m$$

Amt. owing after 48 payments $\times 6$

$$30000(1.011)^{48} - m(1 + 1.011 + \dots + 1.011^{47})$$

$$30000(1.011)^{48} = \frac{m \times 1(1.011^{48} - 1)}{1.011 - 1}$$

$$7119.97 \dots = 62.787m$$

$$m = 807.8060$$

Repayments are \$807.81

Question 9

a) $3x^2 + 5x - 2 = 0$

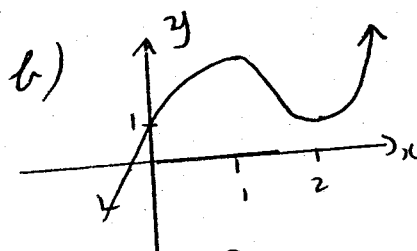
i) $\alpha + \beta = -\frac{5}{3}$

ii) $\alpha\beta = -\frac{2}{3}$

iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{(\alpha\beta)^2}$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{\frac{25}{9} + \frac{4}{3}}{\frac{4}{9}} = \frac{37}{9} \times \frac{9}{4} = \frac{37}{4}$$



c) $V = \pi \int_0^2 (e^x + 1)^2 dx$

$$= \pi \int_0^2 e^{2x} + 2e^x + 1 dx$$

$$= \pi \left[\frac{e^{2x}}{2} + 2e^x + x \right]_0^2$$

$$= \pi \left[\frac{e^4}{2} + 2e^2 + 2 - \left(\frac{1}{2} + 2 \right) \right]$$

$$= \pi \left[\frac{e^4}{2} + 2e^2 - \frac{1}{2} \right] u^3$$

d) $\frac{1}{2}(5x-1) = 4 - \frac{1}{3}(x-2)$

$$3(5x-1) = 24 - 2(x-2)$$

$$15x - 3 = 24 - 2x + 4$$

$$17x = 31$$

$$x = \frac{31}{17}$$

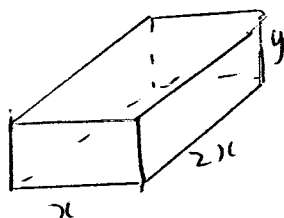
Question 10

Volume = 4

$$x \times 2x \times y = 4$$

$$2x^2 y = 4$$

$$y = \frac{4}{2x^2} = \frac{2}{x^2}$$



$$\text{Cost} = (2 \times 2x^2) \times 15 + (2 \times 2x \times y + 2 \times x \times y) \times 10$$

$$\begin{aligned} C &= 15 \times 4x^2 + 6xy \times 10 \\ &= 60x^2 + 6x \times \frac{2}{x^2} \times 10 \\ &= 60x^2 + \frac{120}{x} \end{aligned}$$

1) For cheapest cost,

$$\frac{dC}{dx} = 0 \quad \text{ie} \quad 120x - \frac{120}{x^2} = 0$$

$$\begin{aligned} 120x &= \frac{120}{x^2} \\ x^3 &= 1 \\ x &= 1 \end{aligned}$$

$$\frac{d^2C}{dx^2} = 120 + \frac{240}{x^3} > 0$$

$\therefore$ Minimum cost when $x = 1$

$\therefore$ Measurements are

1m by 2m by 2m.

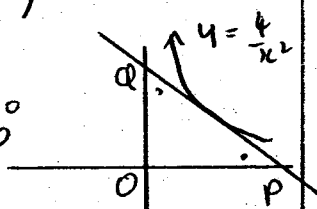
b) $\angle OPQ = \angle OQP$ (given)

$$\angle OPQ + \angle OQP + \angle QOP = 180^\circ$$

($\angle$ sum of Δ)

$$\angle OPQ = \angle OQP = 45^\circ \quad \text{as } \angle QOP = 90^\circ$$

$\therefore$ QP makes an $\angle$ of 135° with the +ve x axis so $m_{PQ} = \tan 135^\circ = -1$



$$\text{ii) } m_{\text{tangent to } y = \frac{4}{x^2} \text{ is } -1$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= -1 \quad \text{ie} \quad -\frac{8}{x^3} = -1 \\ 8 &= x^3 \\ x &= 2 \end{aligned}$$

When $x = 2$, $y = 1$ so the tangent PQ touches the curve at $(2, 1)$

iii) Equation is $y - 1 = -1(x - 2)$ ie $x + y - 3 = 0$

$$\text{c) } y = \cot x = \frac{\cos x}{\sin x}$$

$$\frac{dy}{dx} = \frac{\sin x \times -\sin x - \cos x \times \cos x}{(\sin x)^2}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$